YUANBO PANG

Agent Evaluation & Post-Training Measurement

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Summary

Co-first author of **Agents' Last Exam** (ALE): I build the task environments, automated evaluators, and verifiable-reward pipelines used to measure frontier agents, and run those evaluations with frontier model teams — analyzing trajectories, investigating surprising results, and reporting failure modes back to the labs.

Education

University of California, Berkeley — B.S. EECS, GPA 3.762/4.0 (in-major 3.811) *Expected Dec 2026*

Stanford University (Summer Session) — CS 229: Machine Learning *Summer 2025*

Inspire Fellowship, rednote (Xiaohongshu) — $100K, one of 6 selected worldwide *2026*

Experience

Research Assistant, Berkeley RDI / BAIR (Prof. Dawn Song) *Oct 2025 – Present*

Agents' Last Exam (ALE) — co-first author arXiv:2606.05405 agents-last-exam.org

- Led the task pipeline and auto-scoring for ALE's 1K+ expert-designed, long-horizon tasks (55 subdomains), each an executable environment (`load`/`start`/`evaluate`) with programmatic graders — cross-domain evaluators producing **verifiable rewards** for both measurement and RL post-training (RLVR), with checks that keep graders trustworthy as tasks and models change.
- Operate large-scale sandboxed evaluation on GCP (container/VM runners, incl. computer use); run ALE evaluations with frontier model teams (OpenAI, Anthropic, Meta TBD Lab, Qwen, Seed) — analyzing trajectories and reporting failure modes back to the labs.
- Lead the SWE track (ALE-Replica, first author): hidden suites scored against **real-software oracles** the agent never sees; negative controls force shortcuts to zero. Task production runs as a multi-agent factory — builders author, separate reviewers gate — tracked across a seven-stage pipeline.

Robotics Research, Prof. Masayoshi Tomizuka's Lab, UC Berkeley *2026 – Present*

Large-scale robot-learning data benchmark — advised by Zhuo Xu and Prof. Tomizuka

- Built `roboqa`: dataset-quality benchmark at EgoDex scale (338K episodes, 829 h) — health, diversity, and trainability scored separately; every score published with full provenance.
- Agent workflows inside the data pipeline: VLM-judged annotation audits; raw captures auto-converted to a canonical, embodiment-tagged format.
- Train and post-train VLAs on 500 h of curated data; score the resulting policies in simulation.

Equile — The Dry Lab for AI Scientists (independent side project) *Apr 2026 – Oct 2026*

- Built reproducible computing environments where research agents run real computational studies, each result carrying a provenance record — the same problems as operating an eval fleet: reproducibility, scheduling, monitoring, trustworthy outputs.

Publications

Agents' Last Exam. Yiyou Sun[*], Xinyang Han[*], Weichen Zhang[*], **Yuanbo Pang**[*], et al., Dawn Song. *Under review at NeurIPS 2026.* arXiv:2606.05405 #1 Paper of the Week, Hugging Face Daily Papers (Jun 2026). ([*]co-first author)

ScienceIDE: Turning World's Scientific Codebase into Agent Learnable Environments. Hejia Geng[*], Zesen Huang[*], …, **Yuanbo Pang**, et al. arXiv:2609.19134 #2 Paper of the Day, Hugging Face Daily Papers (Sep 2026).

ALE-Replica: Last Exam for Agents in Reconstructing Professional Software. Yuanbo Pang (first author), et al. *Under double-blind review at CVPR 2027* — agents rebuild professional software from docs; state + GUI checked after every operation.

Agent's GameDev Exam. Yuanbo Pang (2nd author), with Dawn Song's group. *Under double-blind review at ICLR 2027.* Author versions available upon request.

Skills

Evaluation: benchmark & task design, automated evaluators/verifiers, verifiable rewards (RLVR), agent trajectory analysis, sandboxed task environments, regression investigation

Engineering: Python (primary), PyTorch, TypeScript, Docker, GCP, Modal, Postgres, agent harnesses, tool-use & MCP, Linux